

Adversarial Vulnerability of Machine-Learning GNSS Spoofing Detectors to Physically Realizable Attacks

Title Page

Authors

1. Joel Segun Ojerinde (Corresponding author)
Regional Centre for Space Science and Technology Education in Asia and the Pacific (RCSSTEAP), Hangzhou International Innovation Institute, Beihang University, Hangzhou 311115, China
Email: joelojerinde@buaa.edu.cn
2. Wasiu Akande Ahmed (Corresponding author)
Regional Centre for Space Science and Technology Education in Asia and the Pacific (RCSSTEAP), Hangzhou International Innovation Institute, Beihang University, Hangzhou 311115, China
Email: ahmedwasiu24@buaa.edu.cn
3. Akinyinka Olukunle Akande
Federal University of Technology, Owerri, Nigeria
Email: olukunle.akande@futo.edu.ng

Corresponding authors: Wasiu Akande Ahmed, ahmedwasiu24@buaa.edu.cn; Joel Segun Ojerinde, joelojerinde@buaa.edu.cn

Funding

The authors received no specific funding for this work.

Authors' contributions

J.S.O. conceived the study, developed the methodology, implemented the experiments, analysed the results and wrote the manuscript. W.A.A. supervised the work and reviewed the manuscript. A.O.A. reviewed the manuscript. All authors read and approved the final manuscript.

Acknowledgements

We thank the University of Texas at Austin Radionavigation Laboratory for the Texas Spoofing Test Battery recordings and the Finnish Geospatial Research Institute for the open-source FGI-GSRx software-defined receiver used in this study.